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Quantification of the impact of partial replacement of traditional cooking fuels by biogas on global warming: Evidence from Vietnam

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ABSTRACT

Biogas production in small-scale plants is considered to be an appropriate technology for energy supply to rural households. It is being beneficial to the environment through replacement of traditionally used cooking fuels (fuelwood and liquified petroleum gas) and so possibly resulting in the reduction of global warming potential. The problem of biogas loss due to leakage or intentional release results in direct methane emissions, threatening to neutralize the environmental benefits of biogas use. The objective of this study was to quantify the impact of partial replacement of traditional cooking fuels by biogas from two most prevalent sizes of biogas plants (BGPs) (6 m^3 and 9 m^3). The two-phase data collection performed in central Vietnam included semi-structured interviews with biogas plant owners, masons and biogas facilitators. The quantification of Global Warming Potential (GWP) from small-scale biogas plants accounted for four different scenarios: 1. *Original scenario* (before biogas implementation), 2. *Current state scenario* (after biogas implementation with reported 40% loss of biogas), 3. *Zero-loss scenario* (after biogas implementation without biogas loss) and 4. *Optimal scenario* (the exclusive use of biogas). Currently, biogas covers over 60% majority of cooking energy mix of surveyed households. Households' consumption of traditional fuels (fuelwood and liquified petroleum gas) was reduced on average by 58% for fuelwood and 65% for LPG. In comparison of Original and *Current state scenario*, GWP was reduced on average by 11%, suggesting that in the current state, reduction of GWP due to biogas use is under its potential. Two potential scenarios (Zero-loss and Optimal) resulted in the significant GWP reduction, however their achievement is rather theoretical. The break-even point of biogas loss was determined at 53–55%. The findings illustrate the unfulfilled potential of GWP reduction by using biogas and emphasize the problem of significant biogas loss, compromising the environmental benefit of biogas plants.

1. Introduction

Biogas technology has been widely extended in rural areas of South East Asia, where the rapid growth of population resulted in higher demand for meat and development of livestock production, leading to increased amounts of animal manure to be treated (Vu et al., 2012). The number of small-scale biogas plants (BGPs) used for cooking activities is expected to be around 50 million globally, with the majority of digesters built in Asia, including 41.9 million BGPs in China, and 4.9 million in India (Sarika Jain, 2019). The number of BGPs is expected to grow, as the technology has not yet fulfilled its potential and development of household biogas projects is widely supported by policy and subsidization (Wang et al., 2016). The tradition of using biogas in Vietnam comes from 1960s, however a more significant expansion occurred in 1990s and the following years (Nguyen, 2011). In Vietnam, there are currently over 0.5 million small-scale biogas plants (BGPs) (Khan and Martin, 2016), still not reaching the potential of available livestock waste (Anh, 2016). The extension of biogas plants in Vietnam has been supported by a national biogas programme in donor-based cooperation with development organisations (i.e. SNV (Stichting Nederlandese Vrijwilligers), a non-profit international

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development organisation) (Ghimire, 2013).

Small-scale BGPs use on-farm available feedstock, including animal manure and human excreta (Rajendran et al., 2012). In the target area of central Vietnam (Thua Thien Hue province), the main feedstock for the digesters is pig slurry, usually complemented with chicken or other animal manure and human excreta (Roubík et al., 2018). The designs of BGPs used in the target area are KT1 and KT2, with a minimum volume of 4 m^3 (Roubík et al., 2018); however, the survey focused on BGPs with volumes of 6 m^3 and 9 m^3 prevailing in the study area. The choice of a digester size is primarily based on the initial number of pigs in the household, as BGPs serve mainly as a waste management tool. The primary product – biogas - is distributed to the biogas stoves and lamps, the secondary product – digestate - is used as an organic fertiliser in agriculture or as a nutritional source in aquaculture (Nhu et al., 2015). Even though, Maroušek et al. (2020a) highlighted that long-term application of digestate into arable land may result in decline of soil fertility. However, as the study by Prokopchuk et al. (2018) emphasized, agricultural sector is a highly sensitive to natural and climatic, technogenic, anthropogenic, socio-economic and other risks. As this study describes the further importance of insurance coverage in agrarian sector, it is important to note that this service is still underdeveloped in Vietnam, especially regarding poor households (Dao and Tai, 2014). Implementation of biogas in rural households results in the replacement of traditionally used fuels including fuelwood, liquified petroleum gas (LPG) and coal (Thien Thu et al., 2012).

Biogas technology is capable of addressing many problems at the same time, including constantly growing energy demand and management of organic waste resulting from increasing livestock production. In the long term, the technology can contribute to the economic situation of households by reduction of fuel expenses, especially if the initial cost of purchase of a BGP is co-financed by supporting programmes. The paper by Bierer and Götze (2012) discusses various implications for energy cost accounting, it is important to account for energy consumption, energy deliver cost as well as for energy losses to identify cost saving potentials. Furthermore, the use of biogas results in improvement of health and sanitation due to manure treatment by digestion and reduction of indoor air pollution by reduction of the amounts of combusted traditional fuels (Roubík and Mazancová, 2016). The conclusion of the study by Urbancová (2013) emphasizes that innovations and adequate knowledge are prerequisites for successful development. This conclusion can be transferred to BGP, where it is necessary to focus on the training of technology providers and users and at the same time adapt to the ever-changing conditions. As Maroušek et al. (2018) described, there are processes within biogas technology such as steam explosion technology, capable of further improvement of biomass degradation. However, there are different drawbacks as increase of water-consumption or investment cost, causing such improvements to be unfeasible for use at small-scale BGPs in rural areas.

The potential of GHG mitigation is the foremost environmental benefit of small-scale BGPs. Mardoyan and Braun (2015) pointed out, that environmental benefits of biogas use can be highly compromised due to donations on purpose-grown phytomass. This problem does not affect BGPs in South East Asia that much as they are primarily fed with organic waste. The study by Bailis et al. (2015) estimates, that unsustainable wood harvesting and combustion contributes approximately to 1.9–2.3% of global GHG emissions, considering South America and South Asia as the hotspots of fuelwood exhaustion. As reflected in Maroušek et al. (2020), there is option for organic waste to be charred via waste heat to provide low-cost biochar, which can be further technological enhancement with environmental advantages. Streimikiene and Mikalauskiene (2016) concluded, it is important to further develop sustainable energy in favour to meet nowadays energy requirements as well as not to compromise the energy needs of future generations. Excessive consumption of fossil fuels is one of the greatest problems of the present time, with a strong impact on global warming. Oláh et al. (2017) highlighted that factors such as fossil fuel price fluctuation are often likely to result in affection of food prices. This further emphasizes the benefits of biogas use, specifically in form of increasing energy independence, while using on-farm-available waste to energy production. Biogas technology, using on-farm generated organic waste is therefore considered as one of the key solutions of this problem.

After a BGP implementation, generated biogas partially replaces traditionally used fossil fuels or fuelwood in daily cooking activities and BGPs help to reduce emissions from animal manure management (Pei-dong et al., 2007). As traditional fuels regularly remain to be used as complementary energy sources (Roubík et al., 2018), the extent of replacement is one of the key parameters for emission savings. While the composition of fuel mix changes after biogas implementation, the energy requirement of daily cooking activities is not affected by fuel replacement. The energy required for cooking is based on family size, diet and cooking time, regardless of the type of fuel used. Biogas from small-scale BGPs in the target area of this study is used only for cooking without any other application. Research presented in this paper presents a realistic estimation of the extent of biogas replacement, based on the assumption of constant energy requirements. In this study, the calculation of the average amount of energy used for cooking in households and fuel mix ratio before and after implementation of a BGP were conducted for the further basis of calculations of GHG emissions resulting from cooking activities.

The benefit of GHG emission reduction may be compromised due to technical constraints and inappropriate operation of BGPs. A frequent problem of BGPs is biogas leakage through leaks from the digester and the piping system. Additionally, an intentional release is a frequent way of dealing with excessive amounts of biogas in the digester. Resulting biogas loss causes direct methane emissions to the atmosphere (Bruun et al., 2014). The composition of biogas in the target area of Thua Thien Hue province is 64.57–65.44% of methane and 29.31–29.93% of carbon dioxide (Roubík et al., 2018). Although a high methane content is advantageous for energy recovery, it can cause higher damage in case of direct emissions, as methane is a greenhouse gas with global warming potential (GWP) 25 times greater than that of carbon dioxide (Kotas et al., 2001). The extent of biogas loss is estimated to possibly reach 40% of total production (Bruun et al., 2014).

In this study, an assessment of the actual impact of biogas implementation in terms of GHG emissions is performed, as the technical and non-technical constraints related to BGP operation threaten to reduce or outweigh the originally proposed environmental benefits. The quantification of GWP is based on current and potential GHG emissions. The novelty of the paper lies in the development of an aggregated method of GWP assessment using biogas emissions reported through users' self-assessment monitoring (the survey). The objectives of this study were (1) to find the extent of replacement of traditional fuels with biogas, (2) to quantify the reduction of GWP resulting from the partial replacement of traditional fuels with biogas, (3) to find the break-even point of GWP, where the emissions resulting from biogas loss cancel out the emissions avoided by traditional fuels replacement.

2. Materials and methods

Data collection was performed in central Vietnam in Thua Thien Hue province, in five communes (Huong An, Phong Xuan, Huong

Xuan, Phong An, Huong Toan) within two districts (Huong Tra, Phong Dien). The collection was coordinated with local authorities and facilitators (employees of communes, persons responsible for the coordination of biogas projects). The respondents were selected by following criteria: farmers owning and operating small-scale BGP, using biogas on daily basis; masons who received training regarding BGP construction with certification by SNV and participated in BGP implementation; facilitators who participated in a selection of farmers (beneficiaries) for the project and provided them with training regarding biogas technology. During the visits, semi-structured interviews were conducted among three target groups: farmers ($n = 22$), masons ($n = 4$) and facilitators ($n = 5$). The dataset was extended with results previously assembled interviews conducted with farmers ($n = 123$). The interviews focused on the overall performance of biogas plants, use of traditional fuels and problems with biogas technology.

In the study, the average parameters of the BGPs were applied based on the field survey in the study area. These are; a Chinese type of a BGP (KT1 and KT2) of volume 6 m^3 ($n = 72$) and 9 m^3 ($n = 50$) (Fig. 1), feedstock – pig manure mixed with water in the ratio of 1:14 producing on average 1.32 (standard deviation $\pm$ 0.29) m^3 of biogas available for cooking (maximum leakages of 40% included). Hence, the assumed average lifespan biogas production of a BGP is 9646 m^3 (with the assumption of a BGP lifespan of 20 years, regular maintenance of a biogas digester once in 6 year and annual temperature fluctuations). The aim is to demonstrate the average results for the study area rather than to focus on possible individual nuances.

2.1. Calculation of fuel use ratio and daily energy cooking budget

The calculations of fuel use ratio and daily energy cooking budget were based on the assumption of the constant overall requirement of cooking energy before and after biogas introduction, as shown in equation (1),

$$B_{LPG} + B_{FW} = A_{LPG} + A_{FW} + A_{BG} \quad (1)$$

Where variables represent energy from different fuels delivered to the heating surface during cooking activities:

B_{LPG} [MJ] is energy from LPG before the implementation of biogas;

B_{FW} [MJ] is energy from fuelwood before the implementation of biogas;

A_{LPG}, [MJ] is energy from LPG after implementation;

A_{FW} [MJ] is energy from fuelwood after implementation;

A_{BG} [MJ] is energy from biogas after implementation.

For this purpose, amounts of combusted fuels (in kg) were calculated based on households' annual expenses for traditional fuels and their purchase price per kg (both collected from the survey).

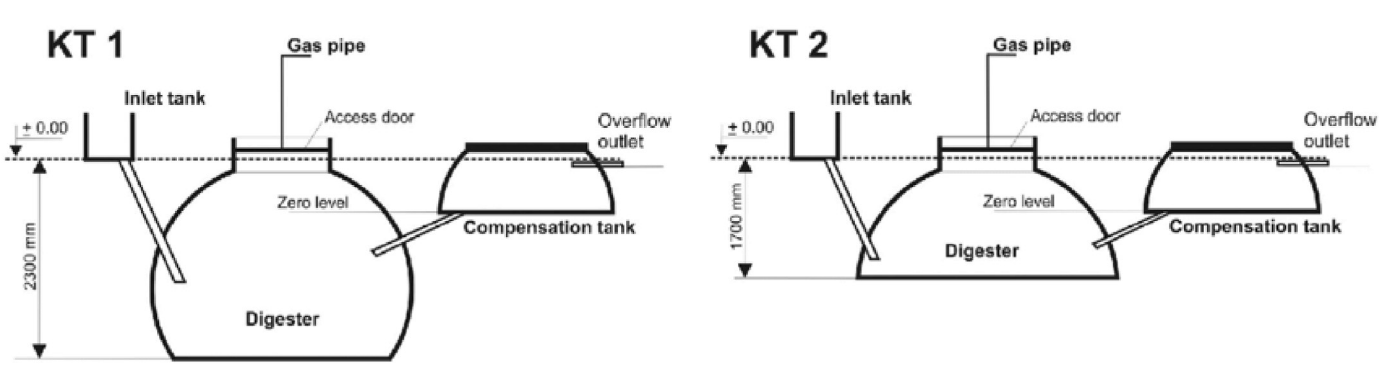

Fig. 1. Scheme of KT1 and KT2 designs.
Source: Roubík et al., (2016).

Table 1
Different scenarios of GHG emissions.

Scenario	Fuels used[a]	Biogas used	Biogas loss [%]
Original scenario	FW, LPG	No	NA
Zero-loss scenario	FW, LPG, BG	Yes	0
Current state scenario	FW, LPG, BG	Yes	40
Optimal scenario	FW, LPG, BG	Yes	0

[a] FW = fuelwood, LPG = liquefied petroleum gas, BG = biogas.

2.2. Calculation of greenhouse gas emissions

GHG emissions were calculated for four different scenarios of fuel use in favour to obtain comparable results for both BGPs volumes of 6 m^3 and 9 m^3. As indicated in Table 1, *Original scenario* accounts for direct emissions of traditional fuels in the state before the biogas introduction, representing the supposedly highest overall GHG emissions. *Current state scenario* accounts for emissions of current fuel use, including biogas loss of 40%, an estimation reported by Bruun et al. (2014). This scenario represents a realistic value of current GHG emissions. *Zero-loss scenario* similarly accounts for emissions of current fuel use ratio, excepting biogas loss. This scenario represents a potential state approachable by proper and uncomplicated operation of a BGP. *Optimal scenario* accounts for a potential state of full replacement of traditional fuels by biogas for cooking purposes.

2.3. Fugitive biogas emissions

Daily biogas yield in the target area was estimated 1.32 m^3 and 1.98 m^3 respectively, based on calculation of model considering volumes of biogas plants of 6 m^3 and 9 m^3. 40% amount of fugitive biogas was based on estimation of Brunn et al. (2014). Calculation of GHG emissions generated during the combustion of LPG, fuelwood and biogas was based on potential emissions of significant GHGs – CO_2, CH_4, CO, N_2O (Table 2), fuel use ratio and daily energy cooking budget.

2.4. Calculation of global warming potential and break-even point

Calculations of GWP [kg CO_2 eqv$\cdot$g^{-1}] resulting from Original, Zero-loss, Current state and *Optimal scenarios* were based on equation (2),

$$GWP = ECB_{CH4}CF_{CH4} + ECB_{CO2}CF_{CO2} + ECB_{CO}CF_{CO} + ECB_{N2O}CF_{N2O} \quad (2)$$

Table 2
GHG emissions per unit of delivered energy.

Emissions of selected gases				
Fuel	CO_2 [g·MJ^{-1}]	CH_4 [mg·MJ^{-1}]	CO [g·MJ^{-1}]	N_2O [g·MJ^{-1}]
Biogas	81.5	57	0.11	5.4
LPG	139	8.9	0.82	6
Fuelwood	532	600	14	4.3

Source: (Rajendran et al., 2012; Siang et al., 2010; Smith et al., 1994; Yu et al., 2008)

Where:

ECB [g·year^{-1}] represents annual GHG emissions released during the combustion of cooking fuels;

CF [g CO2 eqv·g-1] characterization factor of GHG.

The break-even point analysis aims to determine the fraction of biogas loss to occur, in order to obtain an equal GWP, and therefore to negate the positive impact of BGPs to GWP reduction The calculation was based on equation (3)

$$GWP_O = GWP_{ZL} + GWP_{BL} \quad (3)$$

Where:

GWP_O [g CO2 eqv·g-1] represents annual GWP resulting from *Original scenario*;

GWPZL [g CO2 eqv·g-1] represents annual GWP resulting from *Zero-loss scenario*;

GWPBL [g CO2 eqv·g-1] represents GWP resulting from biogas loss.

Supplementary information can be found in Table S1.

3. Results

The survey focused on two designs of BGPs common in the target area: KT1 (36%) and KT2 (64%) with volumes of 6 m^3 and 9 m^3. Majority of the surveyed farmers (91%) planned to continue using biogas technology and reconstruct or build a new BGP after the end of the original lifespan. The family members responsible for BGP operation received training within implementation projects supported by SNV (7 days) or other organisation (1 day). Pig keeping was part of the activity of every small farm, as ownership of at least 5–6 pigs of total body weight of minimum 200 kg was a necessary criterion for inclusion in the programme. Pig manure was the main feedstock for the digester. A minority of households (18%) had toilet connected to the BGP. The reported low share of toilet connection was caused by two reasons such are: the fact that septic was build before the BGP and the perception that the use of human excreta for biogas production is unhygienic.

Every surveyed household was connected to the grid and lighting was supplied by electricity. No use of biogas lamps was mentioned, the only use of biogas was cooking. Majority of the respondents considered biogas production as sufficient (68%). Before biogas implementation, farmers used LPG and fuelwood for cooking. After the implementation, 91% of households used these fuels as complementary sources. Some farmers (9%) reported, that biogas covered their cooking energy consumption in total. After the biogas technology introduction, the consumption of traditional fuels decreased due to the replacement by biogas.

Table 3 shows a comparison of traditional fuels use prior and after biogas implementation, indicating a reduction of their consumption. In the case of 6 m^3 BGPs, the use of fuelwood was reduced by 52%, which resulted in annual savings of 1821 kg of fuelwood. The use of LPG was reduced by 74%, which resulted in annual savings of 179 kg of LPG. In the case of 9 m^3 BGPs, the use of fuelwood was reduced by 63% and the use of LPG was reduced by 56%, Resulting in annual savings of fuelwood and LPG equal to 1975 kg and 117 kg, respectively.

Table 3
Comparison of daily consumption of traditional fuels before and after biogas implementation.

BGP volume	Fuel consumption [kg.d^{-1}]			
	Before biogas implementation		After biogas implementation	
	Fuelwood	LPG	Fuelwood	LPG
6 m^3	9.51	0.66	4.52	0.17
9 m^3	8.58	0.58	3.17	0.26

Based on the mass of combusted fuels, the daily energy mix of cooking activities was calculated for both digester sizes (Table 4). The daily energy budget of cooking activities in case of 6 m^3 and 9 m^3 BGPs was 45 MJ and 40 MJ, respectively.

As indicated in Table 5, in case of 6 m^3 BGPs, biogas took over 61% share in the energy mix after its implementation, while fuelwood and LPG remained to be used, covering the energy requirements from 28% to 10%, respectively. A similar trend has been shown in case of 9 m^3 BGPs, where biogas took over 60% share in the energy mix after its implementation, while fuelwood and LPG remained to cover the energy requirements from 22% to 18%, respectively.

3.1. Greenhouse gases emissions

The annual sums of GHG emissions are listed in Table 6 (for 6 m^3 BGPs) and Table 7 (for 9 m^3 BGPs). In cases of both BGP volumes, scenarios with implemented biogas show reduced emissions of CO_2 and CO in comparison with the *Original scenarios*, which is in accordance with the effort of biogas technology to mitigate GHG emissions. Comparing Original and *Current state scenarios*, emissions of CH_4 increased nearly 15 times in case of 6 m^3 BGPs and 16-fold in case of 9 m^3 BGPs, which occurred as a result of accounted biogas losses.

3.2. Global warming potential and break-even point

Table 8 lists the annual GWPs resulting from different scenarios for BGPs with volumes of 6 m^3 and 9 m^3, indicating that in both

Table 4
Daily amount of energy used for cooking in households with biogas plants of 6 m^3 and 9 m^3.

BGP volume	Daily amount of energy used for cooking in households [MJ.d^{-1}]					
	Before biogas implementation			After biogas implementation		
	Fuelwood	LPG	Biogas	Fuelwood	LPG	Biogas
6 m^3	26.62	18.50	0	4.76	12.65	27.71
9 m^3	24.03	16.18	0	8.88	7.17	24.16

Table 5
Energy mix of daily cooking activities for households with biogas plants of 6 m^3 and 9 m^3.

BGP volume	Composition of energy mix [%]					
	Before biogas implementation			After biogas implementation		
	Fuelwood	LPG	Biogas	Fuelwood	LPG	Biogas
6 m^3	59.00	41.00	0	28.03	10.55	61.42
9 m^3	57.77	40.23	0	22.08	17.83	60.09

Table 6
Annual Greenhouse Gas emissions of selected scenarios (6 m^3 BGPs).

Scenario	CO_2 [kg]	CH_4 [kg]	CO [kg]	N_2O [kg]
Original	6108.34	5.89	141.58	0.41
Current state	3522.40	88.04	67.18	0.55
Zero-loss	3522.40	3.36	67.18	0.55
Optimal	1342.34	0.15	1.81	0.89

Table 7
Annual Greenhouse Gas emissions of selected scenarios (9 m^3 BGPs).

Scenario	CO_2 [kg]	CH_4 [kg]	CO [kg]	N_2O [kg]
Original	5487.00	5.32	127.64	0.73
Current state	2806.30	87.15	48.48	0.77
Zero-loss	2806.30	2.47	48.48	0.77
Optimal	1196.07	0.13	1.61	0.79

cases *Original scenario* results in the highest GWP. Therefore, the state before biogas implementation represents the worst-case of GWP. Compared to the Original state of using only fuelwood and LPG, every scenario implementing biogas resulted in the reduction of GWP.

Current state scenario accounted for 40% of biogas loss, representing a state that can be reasonably projected to occur. Comparison of Original and *Current state scenario* indicates, that in case of 6 m^3 BGPs, the resulting GWP decreased by 10%, and in the case of 9 m^3 BGPs by 13%. *Zero-loss scenario* draws from realistic data about households' energy consumption; however, accounting for 0% of biogas loss can be considered as a barely achievable aim. In this comparison, the reduction of GWP by 41% in case of 6 m^3 BGPs and by 48% in case of 9 m^3 BGPs represents a maximum extent of reduction while still using the current fuel mix. The difference between the GWPs of Zero-loss and *Current state scenarios* embodies a capacity to possible reduction. In the *Optimal scenario* of using biogas exclusively with 0% methane loss, GWP is reduced significantly, by 76% both for 6 m^3 and 9 m^3 BGPs. This represents the best-case GWP scenario achievable by BGP utilization in the conditions of the target area. To approach this scenario, it would be necessary to change the fuel mix ratio to the prevalence of biogas. Preventing biogas leaks to a minimum would also result in a higher amount of biogas to be combusted, therefore generating more energy.

The break-even point analysis was based on GWP of the *Current state scenario* (before biogas implementation) and *Zero-loss scenario*, to which GWP of uprising biogas loss emissions was added as indicated in Figs. 2 and 3. The break-even point was determined at 53% of methane loss for 6 m^3 BGPs and 55% for 9 m^3 BGPs. At the break-even point, the negative impact of biogas production prevails in terms of households' contribution to global warming. This result further emphasizes the dependence of energy generation and emission reduction on prevention of methane loss.

4. Discussion

Results presented in the current study show that the impact of biogas on GWP is a challenge needed to be addressed and at the same time a relevant subject for further research. Despite the large extent of traditional fuels substitution by biogas, which would considerably reduce the GWP of greenhouse gas emissions, there is a problem of biogas loss, which compromises the benefits. Biogas losses are usually caused by leakage from the pipelines, valves or cracks in the gasholder, as well as intentional release in case of unused biogas surplus. The amounts of biogas which are emitted to the atmosphere are problematic to measure, hence in most studies only estimated values are applied. It is presumable, that the fraction of biogas loss greatly varies on each BGP, depending on the used designs and maintenance.

The results show that after biogas implementation covering majority of energy supply, the consumption of traditional fuels decreased significantly. The consumption of fuelwood and LPG was reduced by 58% and 65% on average, respectively. A similar trend was found in a study from Nepal (Katuwal and Bohara, 2009) calculating the reduction of fuelwood by 53% and LPG by 66% with annually saving almost 3 t of fuelwood per household, using small-scale BGPs. The study from Peruvian Andes shows, that fuelwood consumption was reduced by 53%, resulting in annual reduction by 1.88 t per family (Garfí et al., 2012). Chand et al. (2012) calculated, that households using biogas stoves use by 62% less fuelwood in comparison with households using traditional cooking stoves. Izumi et al. (2016) reported, that after implementation of biogas to households in South Vietnam, their consumption of fuelwood dropped by 77% and LPG by 91%, annually saving 2.39 t of fuelwood and 25 kg of LPG per household. It can be concluded, that biogas implementation can result in a large reduction of deforestation.

Zero loss scenario brings the annual GWP decrease by 41% and by 48% for 6 m^3 and 9 m^3 BGPs, respectively. The primary reason for the decrease is a smaller amount of fuelwood combusted, emitting a smaller amount of CO_2. Looking at *Current state scenario*, which considers the worst-case biogas loss scenario with a large amount of biogas (40%) released to the atmosphere, results show the annual decrease of the GWP by 10% and 13%, for 6 m^3 and 9 m^3 BGPs, respectively. This relatively small decrease of GWP shows the seriousness of biogas loss problem. Zero loss scenario can be considered as a confirmation of the great potential for reduction of GWP by biogas utilization. *Current state scenario* in both cases indicates, that the outcome in practice may be far from reaching the potential. The research by Garfí et al. (2012) presents, that after the implementation of biogas in Peruvian Andes, annual GWP of the household was reduced by 50%, resulting in savings of 2704 kg CO_2

Table 8
Annual Global Warming Potential of selected scenarios for biogas plants of 6 m^3 and 9 m^3.

Scenario	GWP from 6 m^3 BGPs [kg CO_2 eqv·g $^{-1}$]	GWP from 9 m^3 BGPs [kg CO_2 eqv·g $^{-1}$]
Original scenario	6767.40	6078.17
Current state scenario	6101.60	5305.08
Zero-loss	3984.58	3188.05
Optimal scenario	1611.83	1436.19

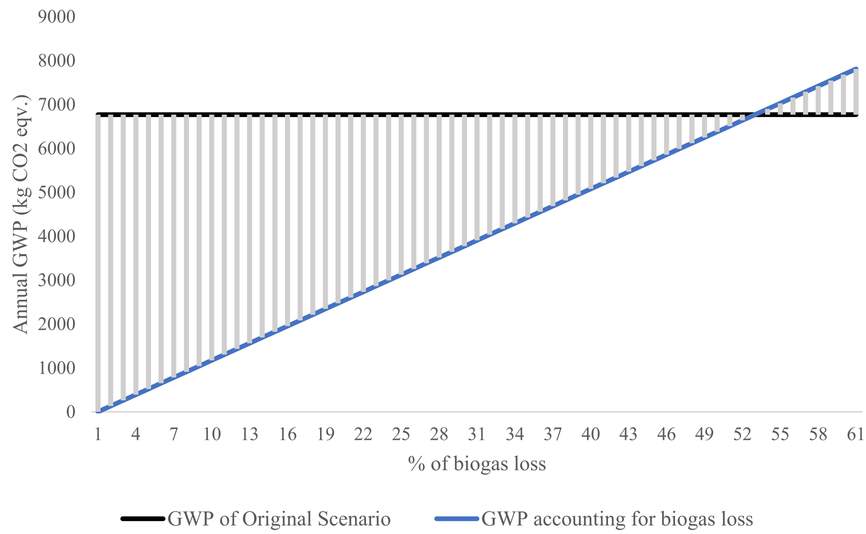

Fig. 2. Break-even point of Global Warming Potential resulting from biogas loss of 6 m^3 BGPs.

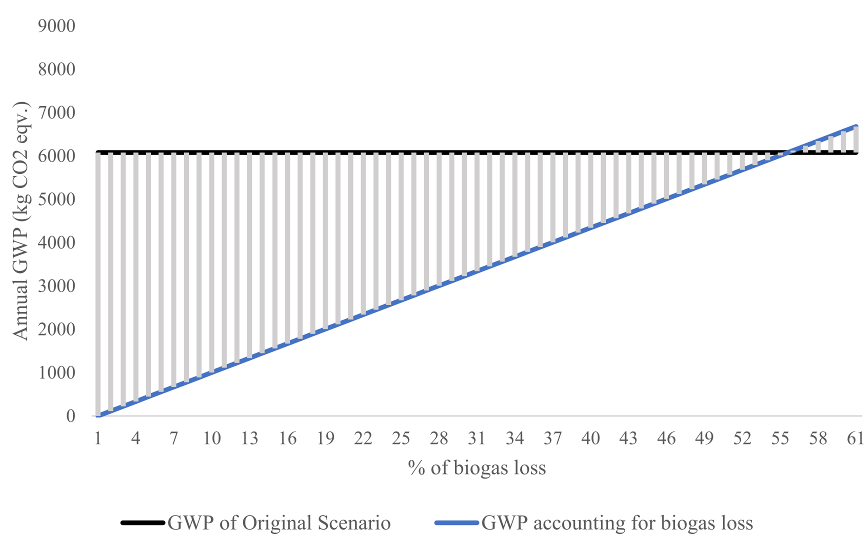

Fig. 3. Break-even point of Global Warming Potential resulting from biogas loss of 9 m^3 BGPs.

eqv$\cdot$g $^{-1}$ due to reduction of fuelwood use and improved manure management. However, this research does not mention any setbacks caused by biogas losses. Hou et al. (2017) performed a study in two target areas in China, along with addressing the problem of biogas loss. The study resulted in contrasting annual values of GWP, with 2668 kg CO_2 eqv$\cdot$g $^{-1}$ increase of GWP in case of poorly managed BGPs in north China and 6336 kg CO_2 eqv$\cdot$g $^{-1}$ decrease in south China on the other hand. The biogas loss, in this case, was calculated by estimation of production and actual consumption. According to Izumi et al. (2016), the annual reduction in emissions resulting from cooking, while replacing fuelwood and LPG was calculated as 1.87 t CO_2 eqv$\cdot$g $^{-1}$. Zhang et al. (2013) calculated the substitution of traditional energy sources by biogas considering BGP with 8 m^3 digester and 20-year life-cycle and applying hybrid Life Cycle Assessment method. The results account for a reduction of CO_2 emissions by 22.84 t per life-cycle, 1.14 t per year, respectively. Although the results indicate, that the emissions of CO_2 throughout the lifespan of the BGP were reduced by 89%, the calculations did not explicitly include potential emissions of CH_4. Emissions of CO_2 and CH_4 are the most significant, when it comes to BGPs impact on the environment and accounting for biogas loss would presumably extend the results of this study. The study by Luo et al. (2017) focused on village-scale BGPs (80 m^3 digester), suggesting, that certain biogas losses are inevitable. However, the emissions could be rapidly reduced through improved BGPs maintenance and appropriate operation. The study suggests, that flexibility maps should be defined to help to completely use the produced biogas. Furthermore, village-scale BGPs are proposed to be suitable for more controlled operation.

The calculations of GHG emissions and GWP are partly based on the method used by Bruun et al. (2014), and modified to the case study of the target area. The study by Bruun et al. (2014) estimates break-even points for separate fuels, which were found at 16% of biogas replacing LPG and at 44% fuelwood. The study presented in this paper aims to reflect the case of the target area and the respective fuel mix, finding break-even point at 53% of biogas loss for 6 m^3 BGPs and 55% for 9 m^3 BGPs. The research results found in this study provide the support for the conceptual premise of the significance of biogas loss for BGPs environmental impact, building on previous results by Bruun et al. (2014) and further examine the topic at the real case studies.

5. Conclusion

This study aimed to quantify the impact of using small-scale biogas plants in central Vietnam on global warming. Data about households' energy consumption and fuel mix ratio were used to calculate current and potential GHG emissions resulting from cooking activities. These emissions were used to quantify the GWP of different scenarios.

In the surveyed households, biogas covers the majority of daily cooking energy, while traditional fuels remain to complement the households' energy requirements. There is still a big potential for replacing traditional fuels by biogas to a greater extent, under the condition of better BGP management, especially by addressing the problem of biogas loss. Calculation of GWP based on cooking activities in the current state resulted in a reduction by 10–13% in comparison to the original state before BGP implementation.

Such reduction appears to be underachieving the actual potential of global warming effect reduction. The greatest possible GWP reductions were quantifiedin cases of two potential scenarios: *Zero-loss scenario* and *Optimal scenario*. These scenarios can only be approached, under conditions of non-problematic use of BGP, resulting in avoidance of undesired emissions. However, the calculated results emphasize the potential of BGPs to reduce households' environmental impact. Break-even points were found at 53–55%, indicating that the current state of BGP use is far below its potential in terms of global warming mitigation. These results also suggest that the problem of methane loss is crucial and should be addressed, to achieve full environmental benefits of biogas technology. As the environmental benefit is one of the reasons for small-scale BGP implementation, further research focusing on identification of their impact on GHG emissions is recommended. Based on the results of this and other studies, it is recommended to support the improvement of BGP operation, ensure the good technical condition and provide better training to the BGP owners. Despite the focus on the case of Vietnam, the method is applicable in other developing countries where small-scale biogas plants of the Chinese type are in operation. In addition, the users' self-assessment reporting approach shows a promising direction to monitor GWP of small-scale biogas plants in developing countries regularly.

CRediT authorship contribution statement

Marek Jelínek: Methodology, Verification, Investigation, Data collection, Formal analysis, Writing - original draft, Writing - review & editing, Visualization. **Jana Mazancová:** Validation, Data curation, Writing - review & editing, Project administration. **Dinh Van Dung:** Investigation, Data collection, Writing - review & editing. **Le Dinh Phung:** Investigation, Data collection, Writing - review & editing. **Jan Banout:** Validation, Writing - review & editing. **Hynek Roubík:** Conceptualization, Methodology, Validation, Formal analysis, Resources, Data curation, Writing - original draft, Writing - review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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References

Anh, H., 2016. Feasibility assessment of anaerobic digestion technologies for household wastes in Vietnam. J. Viet. Env. 7 (1), 1–8. https://doi.org/10.13141/jve.vol7.no1.pp1-8.

Bailis, R., Drigo, R., Ghilardi, A., Masera, O., 2015. The carbon footprint of traditional woodfuels. Nat. Clim. Change 5 (3), 266–272. https://doi.org/10.1038/nclimate2491.

Bruun, S., Jensen, L.S., Khanh Vu, V.T., Sommer, S., 2014. Small-scale household biogas digesters: an option for global warming mitigation or a potential climate bomb? Renew. Sustain. Energy Rev. 33, 736–741. https://doi.org/10.1016/j.rser.2014.02.033.

Chand, M.B., Upadhyay, B.P., Maskey, R., 2012. Biogas option for mitigating and adaptation of climate change. Ku.Edu.Np 1, 5–9. March. http://ku.edu.np/renewablenepal/images/publications/vol1-2.pdf.

Dao, H.T.T., Tai, L. Van, 2014. Agricultural insurance market development the role of Vietnam Government. International Journal of Economics, Commerce and Management II (9), 1–12.

Garfí, M., Ferrer-Martí, L., Velo, E., Ferrer, I., 2012. Evaluating benefits of low-cost household digesters for rural Andean communities. Renew. Sustain. Energy Rev. 16 (1), 575–581. https://doi.org/10.1016/j.rser.2011.08.023.

Ghimire, P.C., 2013. SNV supported domestic biogas programmes in Asia and Africa. Renew. Energy 49, 90–94. https://doi.org/10.1016/J.RENENE.2012.01.058.

Hou, J., Zhang, W., Wang, P., Dou, Z., Gao, L., Styles, D., 2017. Greenhouse gas mitigation of rural household biogas systems in China: a life cycle assessment. Energies 10 (2), 1–14. https://doi.org/10.3390/en10020239.

Izumi, T., Matsubara, E., Dung, D.T., Ngan, N.V.C., Chiem, N.H., Higano, Y., 2016. Reduction of greenhouse gas emissions in vietnam through introduction of a proper technical support system for domestic biogas digesters. J. Sustain. Dev. 9 (3), 224. https://doi.org/10.5539/jsd.v9n3p224.

Jain, Sarika, 2019. Global Potential of Biogas. World Biogas Association, pp. 1–56.

Katuwal, H., Bohara, A.K., 2009. Biogas: a promising renewable technology and its impact on rural households in Nepal. Renew. Sustain. Energy Rev. 13 (9), 2668–2674. https://doi.org/10.1016/j.rser.2009.05.002.

Khan, E.U., Martin, A.R., 2016. Review of biogas digester technology in rural Bangladesh. Renew. Sustain. Energy Rev. 62, 247–259. https://doi.org/10.1016/j.rser.2016.04.044.

Kotas, T., , et al.Luo, T., Pan, J., Fu, L., Mei, Z., Kong, C., Huang, H., 2006. Reducing biogas emissions from village-scale plant with optimal floating-drum biogas storage tank and operation parameters. In: Chapter 4: Fugitive Emissions. IPCC Guidelines for National Greenhouse Gas Inventories, pp. 312–318. https://doi.org/10.1016/j.apenergy.2017.10.036. Applied Energy, 208(July.

Mardoyan, A., Braun, P., 2015. Analysis of Czech subsidies for solid biofuels. Int. J. Green Energy 12 (4), 405–408. https://doi.org/10.1080/15435075.2013.841163.

Maroušek, J., Stehel, V., Vochozka, M., Maroušková, A., Kolář, L., 2018. Postponing of the intracellular disintegration step improves efficiency of phytomass processing. J. Clean. Prod. 199, 173–176. https://doi.org/10.1016/j.jclepro.2018.07.183.

Maroušek, J., Kolář, L., Stunecký, O., Kopecký, M., Bartoš, M., Bartoš, P., Maroušková, A., Cudlínová, E., Konvalina, P., Šoch, M., Moudrý, J., Vaníčková, R., Vrbka, J., 2020. Modified biochars present an economic challenge to phosphate management in wastewater treatment plants. J. Clean. Prod. 272, 123015. https://doi.org/10.1016/j.jclepro.2020.123015.

Maroušek, J., Bartoš, P., Filip, M., Kolář, L., Konvalina, P., Maroušková, A., Moudrý, J., Peterka, J., Šál, J., Šoch, M., Stehel, V., Strunecký, O., Suchý, K., Vochozka, M., Vrbka, J., Zoubek, T., 2020a. Advances in the agrochemical utilization of fermentation residues reduce the cost of purpose-grown phytomass for biogas production. Energy Sources, Part A Recovery, Util. Environ. Eff. 1–11. https://doi.org/10.1080/15567036.2020.1738597.

Nguyen, V.C.N., 2011. Small-scale anaerobic digesters in Vietnam–development and challenges. J. Viet. Env. 1 (1), 12–18. https://doi.org/10.13141/jve.vol1.no1.pp12-18.

Nhu, T.T., Schaubroeck, T., De Meester, S., Duyvejonck, M., Sorgeloos, P., Dewulf, J., 2015. Resource consumption assessment of Pangasius fillet products from Vietnamese aquaculture to European retailers. J. Clean. Prod. 100, 170–178. https://doi.org/10.1016/j.jclepro.2015.03.030.

Oláh, J., Lengyel, P., Balogh, P., Harangi-Rákos, M., Popp, J., 2017. The role of biofuels

in food commodity prices volatility and land use. Journal of Competitiveness 9 (4), 81–93. https://doi.org/10.7441/joc.2017.04.06.
Pei-dong, Z., Guomei, J., Gang, W., 2007. Contribution to emission reduction of CO2 and SO2 by household biogas construction in rural China. Renew. Sustain. Energy Rev. 11 (8), 1903–1912. https://doi.org/10.1016/j.rser.2005.11.009.
Prokopchuk, O., Prokopchuk, I., Mentel, G., 2018. Index insurance as an innovative tool for managing weather risks in the agrarian sector of economics. Journal of Competitiveness 10 (3), 119–131. https://doi.org/10.7441/joc.2018.03.08.
Rajendran, K., Aslanzadeh, S., Taherzadeh, M.J., 2012. Household biogas digesters-A review. Energies 5 (8). https://doi.org/10.3390/en5082911.
Roubík, H., Mazancová, J., 2016. Small-scale Biogas Sector in Central Vietnam: Ways of Financing of the Technology. Agribusiness and the Rural Economy, October, pp. 312–318. Agrarian Perspectives Xxv: Global and European Challenges for Food Production.
Roubík, H., Mazancová, J., Banout, J., Verner, V., 2016. Addressing problems at small-scale biogas plants: a case study from central Vietnam. J. Clean. Prod. 112, 2784–2792. https://doi.org/10.1016/j.jclepro.2015.09.114.
Roubík, H., Mazancová, J., Phung, L.D., Dung, D.V., Banout, J., 2018. Biogas quality across small-scale biogas plants: a case of central vietnam. Energies 11 (7). https://doi.org/10.3390/en11071794.
Siang, C.C., Energy, N., Group, C., 2010. China ' s medium to long-term renewable energy development plan – promotion of bio-ethanol introduction and future strategic, pp. 1–11. July 2006.
Smith, K.R., Apte, M.G., Yuqing, M., Wongsekiarttirat, W., Kulkarni, A., 1994. Air pollution and the energy ladder in asian cities. Energy 19 (5), 587–600. https://doi.org/10.1016/0360-5442(94)90054-X.
Streimikiene, D., Mikalauskiene, A., 2016. Comparative assessment of sustainable energy development in the Czech republic, Lithuania and Slovakia. Journal of Competitiveness 8 (2), 31–41. https://doi.org/10.7441/joc.2016.02.03.
Thien Thu, C.T., Cuong, P.H., Hang, L.T., Chao, N. Van, Anh, L.X., Trach, N.X., Sommer, S.G., 2012. Manure management practices on biogas and non-biogas pig farms in developing countries - using livestock farms in Vietnam as an example. J. Clean. Prod. 27, 64–71. https://doi.org/10.1016/j.jclepro.2012.01.006.
Urbancová, H., 2013. Competitive advantage achievement through innovation and knowledge. Journal of Competitiveness 5 (1), 82–96. https://doi.org/10.7441/joc.2013.01.06.
Vu, Q.D., Tran, T.M., Nguyen, P.D., Vu, C.C., Vu, V.T.K., Jensen, L.S., 2012. Effect of biogas technology on nutrient flows for small- and medium-scale pig farms in Vietnam. Nutrient Cycl. Agroecosyst. 94 (1), 1–13. https://doi.org/10.1007/s10705-012-9516-y.
Wang, X., Lu, X., Yang, G., Feng, Y., Ren, G., Han, X., 2016. Development process and probable future transformations of rural biogas in China. Renew. Sustain. Energy Rev. 55, 703–712. https://doi.org/10.1016/j.rser.2015.09.097.
Yu, L., Yaoqiu, K., Ningsheng, H., Zhifeng, W., Lianzhong, X., 2008. Popularizing household-scale biogas digesters for rural sustainable energy development and greenhouse gas mitigation. Renew. Energy 33 (9), 2027–2035. https://doi.org/10.1016/j.renene.2007.12.004.
Zhang, L.X., Wang, C.B., Song, B., 2013. Carbon emission reduction potential of a typical household biogas system in rural China. J. Clean. Prod. 47, 415–421. https://doi.org/10.1016/j.jclepro.2012.06.021.